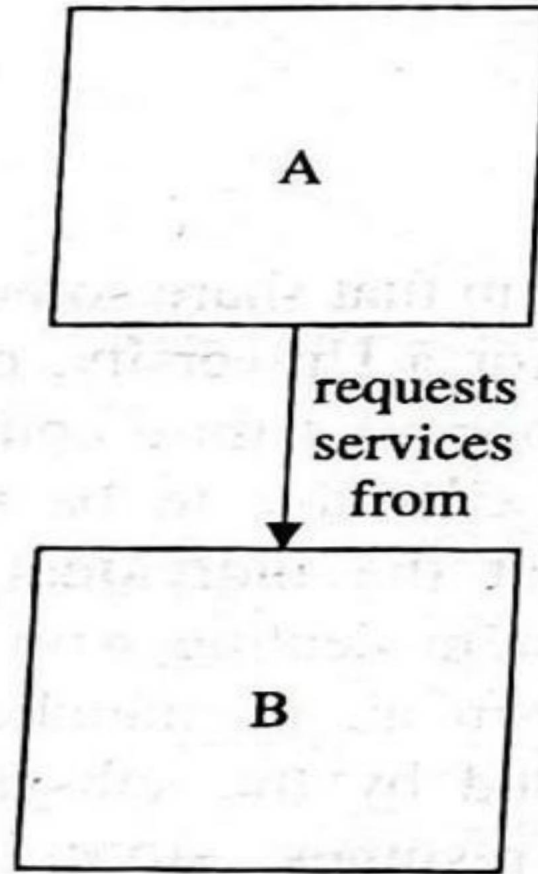


Software Engineering

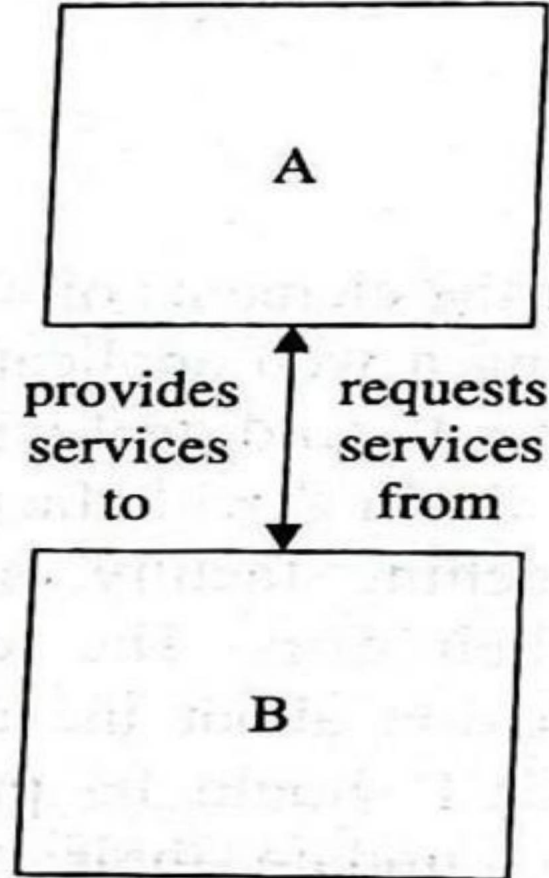
Dr. Md. Nasim Adnan

Architectural Modeling

A software architecture where the client requests services and the server provides those services over a network.



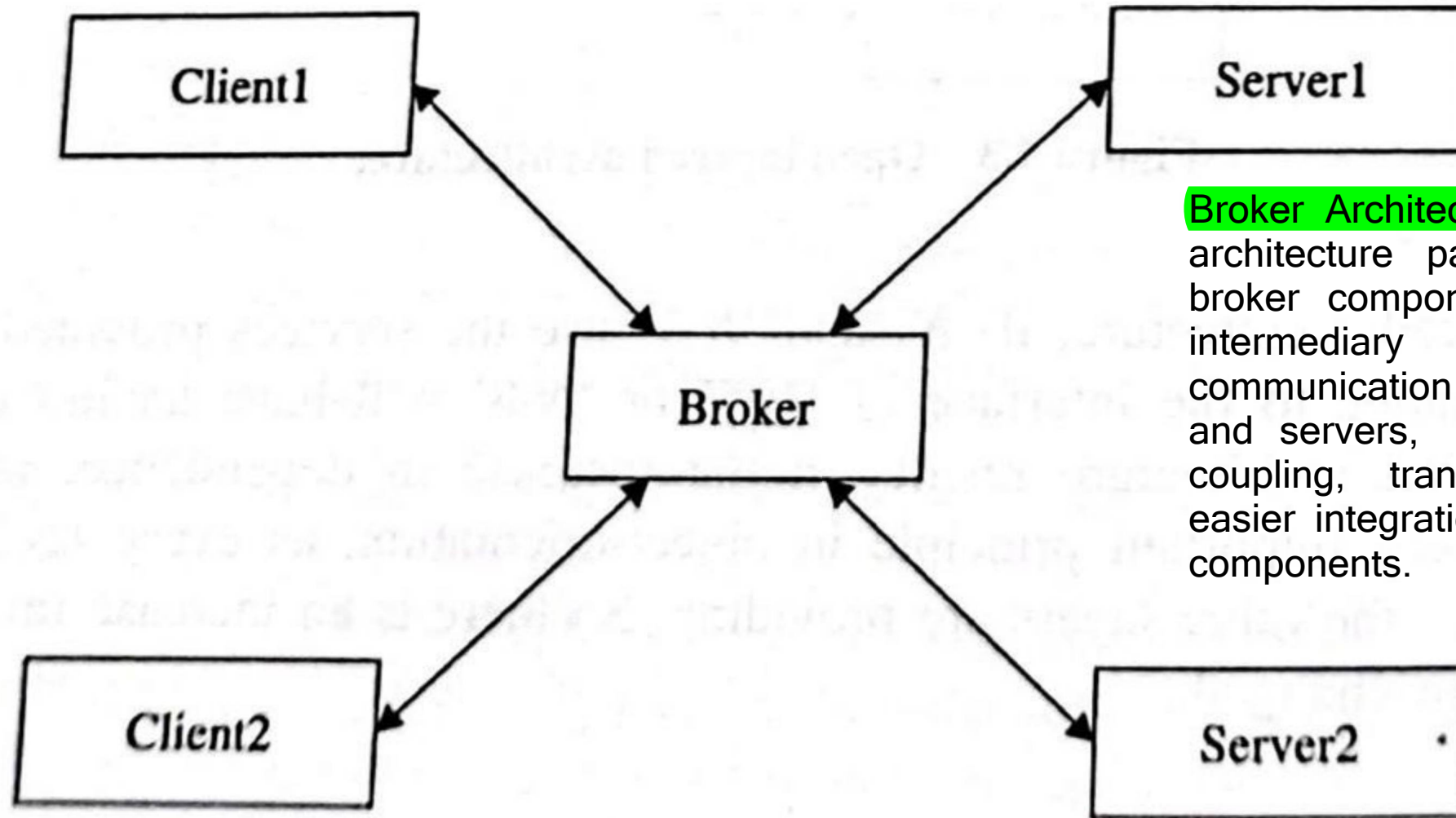
Client-server style



Peer-to-peer style

Peer-to-Peer architecture is a decentralized software architecture where each node (peer) in the system can act as both a client and a server, allowing direct communication and resource sharing without relying on a central server.

A broker is an intermediary that coordinates communication between clients and servers (or different components). Clients don't call servers directly; instead, they go through the broker.



Broker Architecture: A software architecture pattern in which a broker component acts as an intermediary to coordinate communication between clients and servers, enabling loose coupling, transparency, and easier integration of distributed components.

Closed Layer: A layered architecture style where each layer can only communicate with the layer directly below it.

Open Layer: A layered architecture style where a layer can communicate with not only the layer directly below it but also any lower layer.

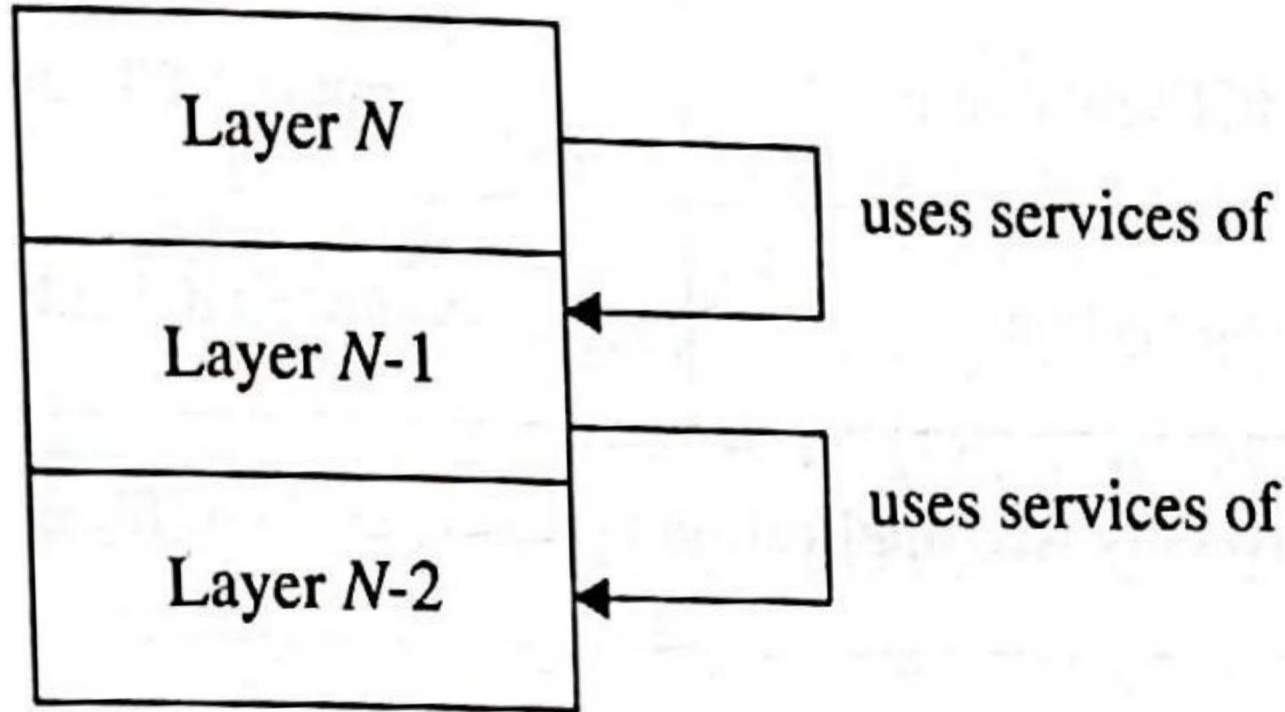


Figure 7.2 Closed layered architecture.

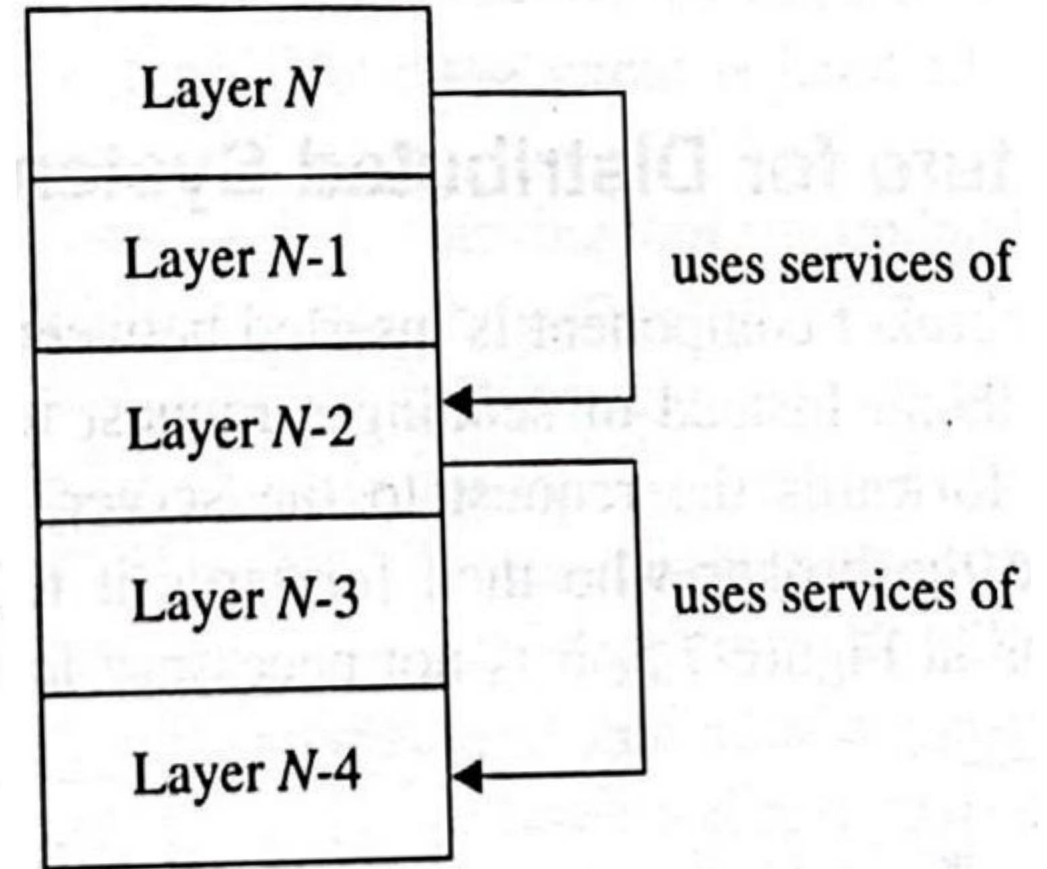
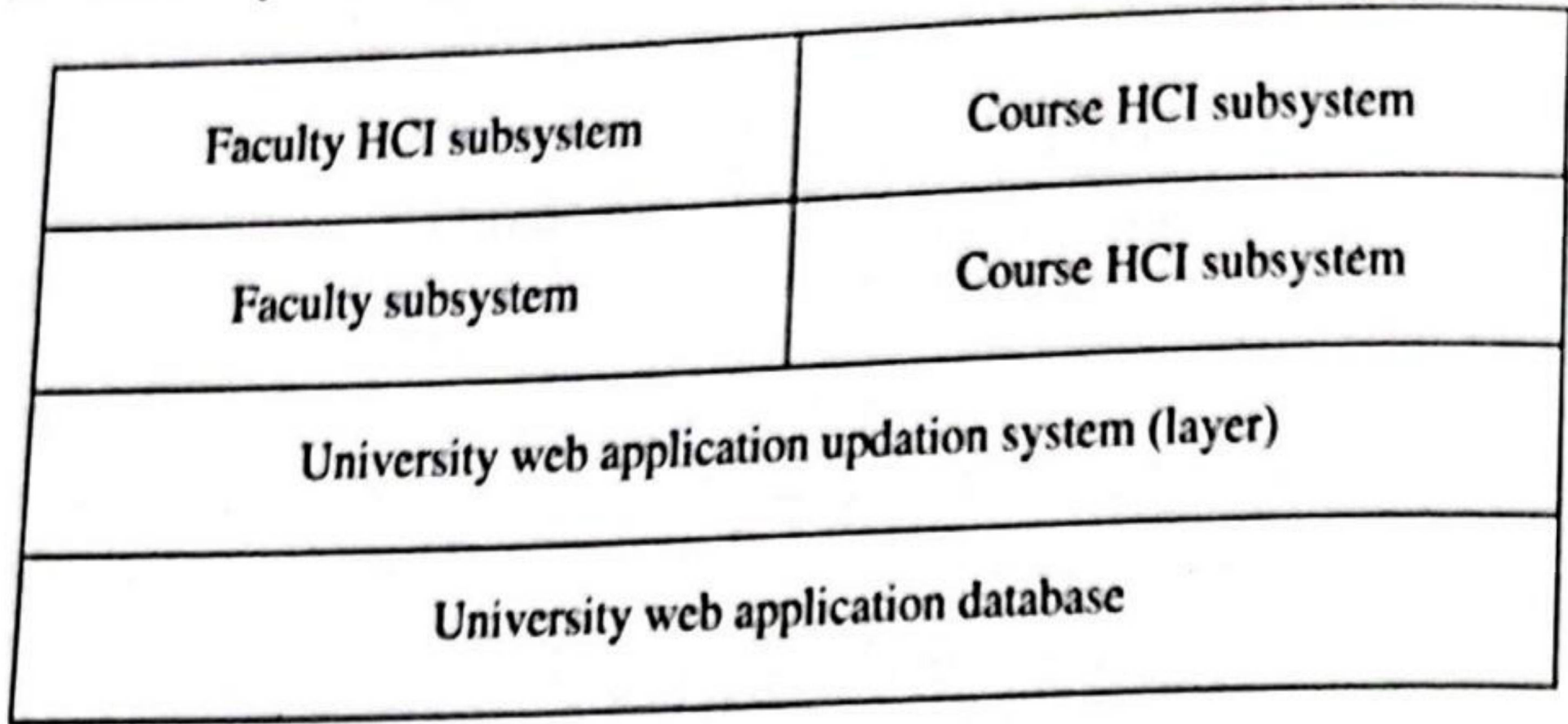


Figure 7.3 Open layered architecture.

Layered Architecture: A software architecture style in which the system is organized into layers, each with a specific role, where higher layers use the services of lower layers, promoting modularity, separation of concerns, and maintainability.

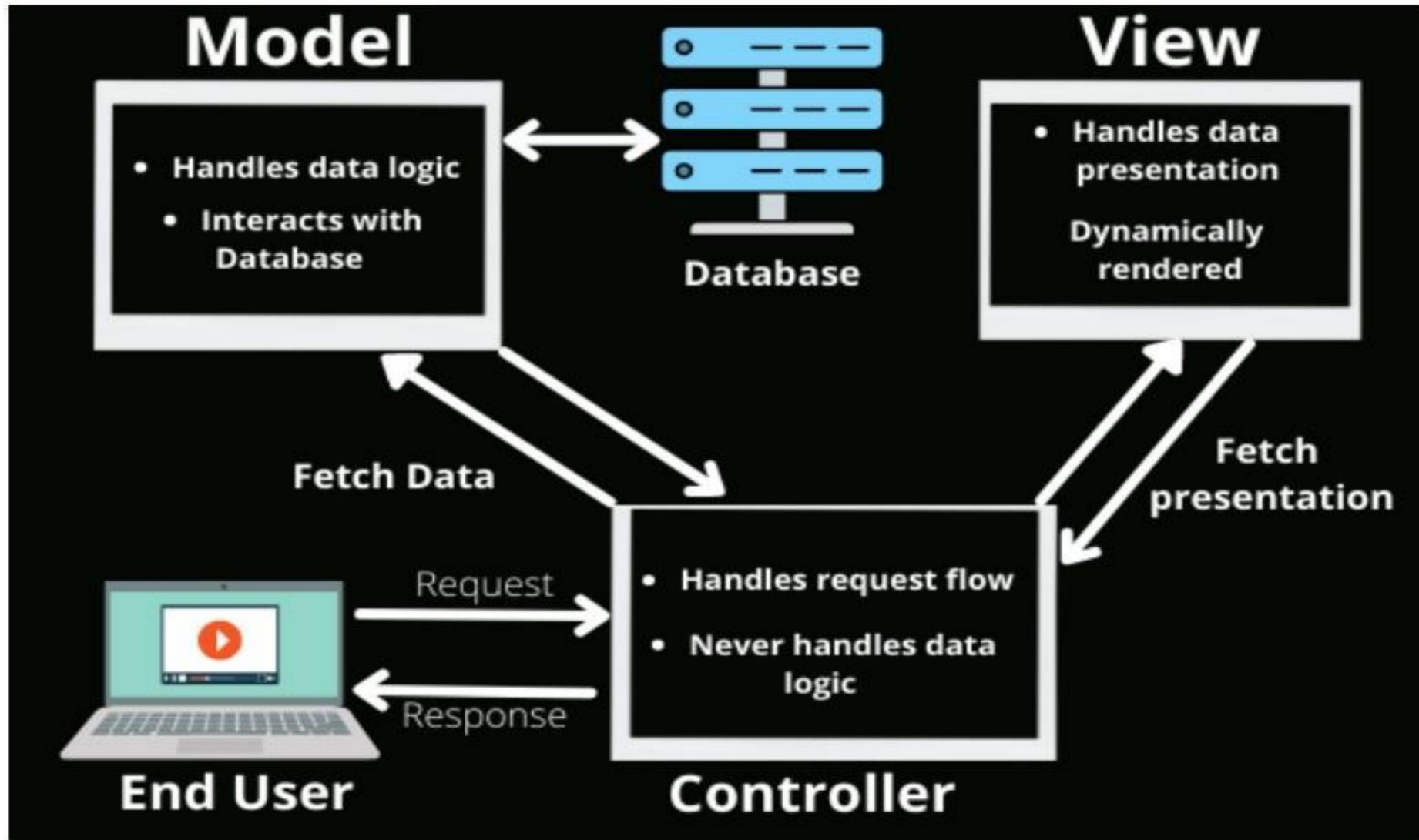
Partitioning



Partitioning: The process of dividing a software system into separate, independent modules or components (partitions) so that each can be developed, maintained, and executed independently, improving modularity, scalability, and manageability.

MVC

- MVC, or Model-View-Controller, is a software Architectural Framework that organizes code into three interconnected components:
 - **Model:** Represents the data and business logic. It handles data retrieval, storage, and manipulation, ensuring that the application functions correctly.
 - **View:** The user interface or presentation layer. It displays data from the model and sends user inputs to the controller.
 - **Controller:** Acts as an intermediary between the model and the view. It processes user input, updates the model, and determines which view to display.



Architectural Framework:

A structure that provides an implementation-specific skeleton for software systems, consisting of reusable design patterns in a defined architecture. It helps improve code organization, scalability, and maintainability by separating concerns and making development more efficient.

Examples: ASP.NET MVC, Django, Ruby on Rails.

